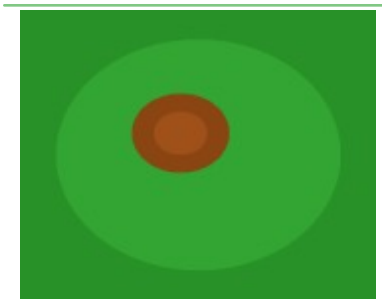
FASAL SURAKSHA (■■■ ■■■■■■■)

AI Micro-Crop Diagnostic, Yield Forecast & PMFBY Verification Engine

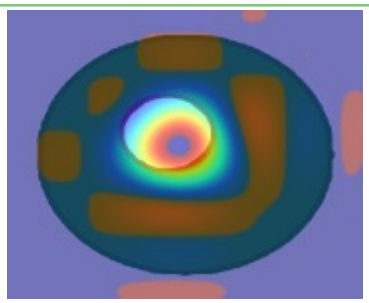
Issue Date: 11 Oct 2026, 19:40
Card ID: FSC-727818
Verification: Verified via Grad-CAM XAI

Target Crop:	Potato	**Diagnosis:**	Early blight	**Severity:**	**2.39% (Mild)**	**AI Confidence:**	96.5%

Original Leaf Capture



Grad-CAM Explainable AI Heatmap



Yield Impact Simulation (XGBoost)

Potential Healthy Yield: 24.0 Tons/Acre
Forecasted Harvest Yield: **15.49 Tons/Acre**
Projected Deficit / Loss: **35.46%**

Model incorporates crop age, seasonal rainfall index, soil pH & pathological leaf necrosis.

Prescribed Treatment Plan (ICAR / Agronomic Standards)

Category	Prescribed Formulation	Concentration & Tank Dose	Frequency & Safety
Organic / Bio	**Neem Oil 10,000 PPM + Trichoderma harzianum** ■■■ ■■■ 10,000 ■■■■■■ + ■■■■■■■■■■■■	• 3.0 mL (or g) / Liter water • 45.0 mL (or g) for 15.0L tank	Every 7 days. Spray at first sign of concentric target spots.
Chemical Systemic	**Mancozeb 75% WP (Dithane M-45)** ■■■■■■■■ 75% ■■■■■■■■■■ (■■■■■■ ■■-45)	• 2.5 mL (or g) / Liter water • 37.5 mL (or g) for 15.0L tank	Every 10 days. Wear gloves.

PMFBY Crop Insurance Evidence & Verification Dossier

Claim Token: PMFBY-20261011-9750	**Farmer / KCC ID:** KCC-UP-98214	**Eligible: YES (Claim Admissible)**	**Estimated Indemnity: INR 53,190.00**
Audit Priority: TIER-2: FAST-TRACK DIGITAL ADJUDICATION (■■■■■ ■■■■■■■■■■)	**Surveyor Advice:** Damage crosses 20% PMFBY statutory threshold. Eligible for algorithmic appraisal cross-verified against block-level Automatic Weather Station (AWS) rainfall logs.	**Metadata & Integrity:** Tamper Score: 96% | Geo: Verified	**Scheme:** PMFBY Localized Calamity

Digitally generated by FASAL SURAKSHA AI Platform. Complies with PMFBY automated claims guidelines and ICAR crop protection standards. For re-scan or dispute redressal, visit nearest Krishi Vigyan Kendra (KVK).